Voltage-Gated Harmonic Topology (VGHT)

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Voltage-Gated Harmonic Topology (VGHT)

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A three-mechanism physical substrate for domain-orthogonal cognition with sparse cross-domain analogical bridging and avalanche suppression, built entirely from scalar field dynamics — no rules, no templates, no domain metadata.

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0. Motivation

The Continuous Morphological Dynamics (CMD) substrate — thermodynamic habituation, byte-wave sensory injection, phase-locked syntax emission — produces coherent, non-templated output but exhibits two coupled failure modes:

1. **Domain leakage.** In the shared-oscillator field, a driven medical concept can transfer energy into unrelated conceptual clusters (e.g., gaming, legal, code) whenever their macro-node phases happen to align. The reader observes a medical prompt producing a legal-sounding word in the verb slot, or the middle word of a coherent triple stepping outside the input's semantic neighborhood.
2. **Polymath rigidity.** The dual failure — with sufficient emission-floor thresholds, the substrate becomes strictly domain-locked, unable to draw analogical parallels between distantly-related concepts even when the graph explicitly supports them through L6-A analogical bridges.

The Continuous Morphological Dynamics literature treats these two failures as separate tuning

problems (emission floor, wavefront gate, cluster-energy depletion). Voltage-Gated Harmonic Topology (VGHT) resolves both under a single physical postulate: *domains are orthogonal frequency bands, and bridges between them are voltage-gated channels that only conduct under simultaneous high-amplitude resonance.* Domain isolation and polymath bridging cease to be competing objectives; they become a single dynamical regime with a single parameter, θ_critical, the tunneling threshold.

The mechanism is inspired by three well-understood physical analogs: mode-orthogonal spectral separation in coupled-oscillator networks, voltage-gated ion channels in biological neurons (in particular the sigmoidal open probability of the NMDA receptor with respect to membrane potential), and the GABAergic global inhibitory tone that suppresses run-away excitation in cortical circuits. None of the three requires an explicit "domain tag" or a rule dispatcher; each is a continuous scalar update evaluated once per tick.

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1. Architecture

Let the cavity population be indexed by `i ∈ [0, N)`, partitioned into macro-clusters `C_1, C_2, ...` by the connected components of the tunnel graph (union-find at bind time; the same clustering used by HNCA). Let a slot layer of `S` phase oscillators sit above the macro layer as in CMD phase-locked emission.

1.1 Domain Sub-Bands

Each macro-cluster `C_k` receives a *base natural frequency* drawn from a discrete band-bank:

ω`_k = `ω`_base(k mod B) + jitter(k)`

where `B` is the number of orthogonal bands (default `B = 6`) and ω`_base` is a fixed evenly-spaced ladder in the range `[`ω`_lo, `ω`_hi]`, e.g. `[8, 32] rad/s` giving band centres at `{8, 12, 16, 20, 24, 28}`. The jitter is a small in-band perturbation (± 0.5 rad/s) drawn deterministically from the cluster id, ensuring reproducibility across seeds while still breaking degeneracy inside a band.

The verb-material ω-boost from prior CMD work becomes an *in-band shift* rather than a cross-band jump:

ω`_k ← (1 − p_k)·`ω`_k + p_k·`ω`_verb_offset(k)`

where `p_k ∈ [0, 1]` is the verb-material share of cluster `k`'s internal edges (`verb_pull_for(rel) / |internal edges|`), and ω`_verb_offset(k)` is the *upper edge* of cluster `k`'s home band — so verb-heavy clusters rise to the top of their band without colliding with a neighbouring band. This preserves both the domain orthogonality of Sub-Bands and the verb-slot phase-lock property from the prior implementation.

Consequence. Two clusters in different bands share almost no common Kuramoto phase drift, because their ω differs by at least a band-width and the coupling terms `sin(`θ`_j − `θ`_i)·A_j` average to zero over any window longer than the beat period. Domain interference during ordinary conversation is suppressed by frequency orthogonality alone — no relation table, no cluster tag, no rule.

1.2 Voltage-Gated L6-A Tunnels

L6-A tunnels are the analogical bridges discovered by structural resonance between cross-cluster pairs (Phase 1.3 online-L6A-discovery of the CMD line of work). In VGHT, every *inter-cluster* tunnel — and every tunnel discovered by L6-A regardless of cluster membership — is marked *voltage-gated*. Its force is scaled at every tick by a sigmoidal conductance:

`C(A_a, A_b) = C_max · σ(κ · (A_a · A_b − θ_critical))`

where `A_a = |x_a|` and `A_b = |x_b|` are the instantaneous displacements of the two endpoint cavities, σ is the logistic function, κ is a steepness parameter (default `12`), and `θ_critical` is the tunneling threshold (default `0.05` in the field's natural units). `C_max = 1`.

The force delivered to endpoints is scaled elementwise by `C`:

`F_a ← F_a · C(A_a, A_b)`
`F_b ← F_b · C(A_a, A_b)`

Consequence. When either endpoint is quiescent, `A_a · A_b ≈ 0`, `C ≈ σ(−κθ_critical) ≈ 0`, and the bridge presents effectively infinite resistance. The two clusters exchange no energy. When both endpoints are simultaneously excited to a level whose product exceeds `θ_critical`, `C → 1` sharply and the bridge conducts fully. This is the resonant/quantum-tunneling analog explicitly identified in the motivating architecture: the door opens only when the room on both sides is loud enough to make the door irrelevant.

Because both endpoints must resonate at once, spurious cross-domain leakage — a stray phase alignment on one side of a bridge — cannot open the channel. Only genuine dual excitation, produced by an input whose structural resonance actually spans two domains, does so. This is the polymath spark.

1.3 Global Viscosity (GABA analog)

The system tracks its total kinetic energy each tick:

`E_total = ½ Σ_i m_i · v_i²`

An effective friction coefficient is derived as a saturating function of `E_total`:

`μ_eff = μ_0 + k_avalanche · (1 − exp(−E_total / E_scale))`

with defaults `k_avalanche = 0.10`, `E_scale = 5.0`. `μ_eff` replaces the constant friction in the cavity velocity-decay step and in the tunnel-damping term of that tick.

Consequence. Under normal, sparse activation `E_total` is small, `μ_eff ≈ μ_0`, and the field evolves as before. Under an avalanche — many nodes firing simultaneously through spurious phase alignment — `E_total` rises rapidly, `μ_eff` saturates toward `μ_0 + k_avalanche`, and the excess kinetic energy is dissipated in the same tick. Only the field components whose amplitudes are large enough to overcome the elevated friction survive; the noise floor is damped out. This is the physical analog of GABAergic inhibitory tone.

Unlike the raw `exp(E_total)` formulation, the bounded `1 − exp(−E_total / E_scale)` form cannot diverge, guaranteeing numerical stability under any input, however extreme.

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2. Interaction of the Three Mechanisms

The three mechanisms are not independent tuning knobs; they are coupled by design.

- **Sub-Bands** determine which cavities *can* phase-lock together in the absence of external excitation. Same-band cavities freely couple; cross-band cavities decouple by frequency alone.
- **Voltage-gated bridges** re-establish cross-band coupling — but only when both endpoints are already excited to a level that represents genuine resonance with the input. Without such excitation, the bridge is inert regardless of the graph's connectivity.
- **Global viscosity** suppresses false excitation. When a spurious cascade begins to raise `E_total`, μ_eff rises and the cascade dies before it can meet the `A_a · A_b > θ_critical`

condition needed to open any bridge.

The joint regime is: *within-band coupling is free and continuous, cross-band coupling is possible but expensive, and pathological cross-band cascades are physically forbidden by dissipation.* Domain isolation and polymath bridging cease to be tradeoffs.

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3. Physical Interpretation

Each of the three mechanisms has a direct biological analog and an independent literature suggesting the same functional role.

1. **Frequency-band orthogonality** appears in cortical cognition as the observation that different cognitive processes (attention, working memory, motor planning) show distinguishable EEG band signatures (α, β, γ). Cortical microcircuits couple preferentially within-band, and cross-band coupling is a marker of deliberate binding (theta-gamma phase-amplitude coupling).
2. **Voltage-gated channels** are the mechanism by which NMDA receptors implement coincidence detection: their conductance is sigmoidal in the product of presynaptic glutamate release and postsynaptic depolarization, mirroring `C(A_a, A_b) =` σ`(A_a · A_b −` θ`)` structurally.
3. **GABAergic inhibition** provides the global tone that keeps cortical excitability in the sparse-firing regime. Loss of GABAergic inhibition produces exactly the avalanche pattern VGHT's viscosity term suppresses: dense synchronous firing across unrelated cortical areas, the neural correlate of a seizure.

VGHT does not claim to be a model of any of these biological mechanisms. It claims that a substrate exhibiting the *dynamical signatures* of all three simultaneously — with only three scalar parameters — provides the minimum architectural surface area on which domain-orthogonal cognition with sparse analogical bridging is a stable regime, rather than an accidental artifact of parameter tuning.

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4. Implementation Contract

- **No if/else branches** in the tick loop that inspect a relation label, a domain tag, or a cluster identifier. All three mechanisms are pure scalar arithmetic on existing per-cavity or per-tunnel fields.
- **No new I/O boundary code** for domain metadata. Sub-Bands are assigned at bind time from cluster id, computed by the union-find that HNCA already runs. Voltage gating is a per-tunnel flag set once at bind for cross-cluster tunnels (and for any L6-A tunnel regardless of cluster membership). Viscosity is derived each tick from the same `total_kinetic` accumulator already maintained for the RIFA IC pass.
- **All defaults ON.** VGHT is not a feature flag. The three mechanisms are the substrate's normal dynamics.

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5. Parameters

Symbol	Default	Meaning
`B`	6	Number of domain sub-bands
ω`_lo`, ω`_hi`	8, 32 rad/s	Band ladder range

Symbol	Default	Meaning
κ	12	Voltage-gate steepness
θ_critical	0.05	Voltage-gate tunneling threshold
k_avalanche	0.10	Peak added friction under avalanche
E_scale	5.0	Kinetic-energy saturation scale

These are the only knobs. All other quantities are derived.

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6. Expected Empirical Signature

A VGHT-enabled substrate is expected to show:

- **Single-domain prompts** produce single-domain phase-lock emissions with no cross-domain leakage into any slot, even after many turns of a single-domain conversation. The wavefront gate and the sub-band orthogonality are jointly responsible.
- **Cross-domain prompts** — where the input drives cavities in two different bands to simultaneous high amplitude — open the voltage-gated bridges between those bands within a few ticks, producing a genuine polymath emission that spans domains.
- **Injection of a single garbage or nonsense token** produces at worst its own auto-spawned label in output; the avalanche suppressor prevents its stray phase drift from recruiting unrelated cavities.
- **Latency to cross-domain emission** is measurable: it should correspond to the time required for the voltage-gate sigmoid to saturate under paired excitation, i.e., a few tens of ticks. This latency is the phenomenological signature of the polymath spark and is measurable with the existing readout infrastructure.

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7. Relation to Prior CMD Work

VGHT subsumes and refines three earlier CMD components without discarding them.

- Phase 2.2 rubber-tunnel band-pass filters are a special case of voltage-gated tunnels with θ_critical = 0 (always conducting) and a Gaussian gate on ω alone. VGHT's sigmoidal amplitude gate is strictly more expressive.
- Phase 3.1 multi-ontology substrates rely on manually loaded disjoint axiom files to achieve domain separation. VGHT achieves the same isolation from a single ontology by frequency orthogonality alone, and further allows analogical bridging without touching the ontology.
- The cluster-energy depletion term in phase-locked emission remains a fallback for the readout stage. When VGHT is fully active, its contribution to output diversity is redundant with the sub-band separation.

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8. Failure Modes

The design has three known failure modes worth documenting explicitly.

1. If the cluster count is less than B, some bands go unused, which is harmless but wastes spectral capacity.
2. If θ_critical is set too high, cross-domain bridges never open even for well-posed polymath prompts. If set too low, the system reduces to the pre-VGHT shared-oscillator field.

3. If `k_avalanche` is set too high, meaningful large-amplitude emissions are damped alongside noise; the system becomes under-expressive. Empirical tuning against a held-out prompt set is suggested.

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9. Summary

VGHT closes the domain-isolation / polymath-bridging tradeoff by mapping domains to orthogonal frequency bands, making cross-domain bridges voltage-gated by joint amplitude resonance, and adding a global energy-dependent viscosity that dissipates avalanches before they can spuriously open bridges. All three mechanisms are scalar physics with a total of six parameters. The tick loop remains free of if/else on relations, domain tags, or cluster identifiers. The result is a substrate that is simultaneously domain-locked in default operation and polymath-capable when its input carries structural resonance across bands — the physical realization of the intuition that a polymath is not a system that mixes domains randomly, but one whose domains are ordinarily silent to each other and speak only when the same underlying pattern excites both.